

Application of Nyquist theorem

- ▶ For each of the oscillatory signals you have generated earlier, find the highest instantaneous frequency in the signal
 - ▶ Example: For a linear chirp with instantaneous frequency $f_0 + f_1 t$, the highest frequency is
- ▶ While this may not hold for all cases, the highest instantaneous frequency in an oscillatory signal is a good approximation to its Fourier domain bandwidth
- ▶ Use the above approximation to sample your signals at
 - ▶ Nyquist rate: $2 \times$ highest frequency
 - ▶ Twice the Nyquist rate
 - ▶ $10 \times$ Nyquist rate
- ▶ Plot the signals in both time and Fourier domains as you have done before

Filtering

- ▶ Use the function for generating a sinusoid to generate a signal containing the **sum of three sinusoids** with the following parameters
- ▶ Number of samples: 2048
- ▶ Sampling frequency: 1024 Hz
 - ▶ What is the maximum frequency of the discrete time sinusoid you can generate with this sampling frequency?

	Signal 1	Signal 2	Signal 3
A	10	5	2.5
f_0	100	200	300
ϕ_0	0	$\pi/6$	$\pi/4$

- ▶ Use Matlab's `fir1` function to design 3 **different** filters such that filter #i allows only signal #i to pass through
- ▶ Apply each filter to the signal and show the periodogram of the input and outputs

FIR filter design

► >> help fir1

`fir1` FIR filter design using the window method.

`B = fir1(N,Wn)` designs an N'th order lowpass FIR digital filter and returns the filter coefficients in length N+1 vector B.

The cut-off frequency W_n must be between $0 < W_n < 1.0$, with 1.0 corresponding to half the sample rate. The filter B is real and has linear phase. The normalized gain of the filter at W_n is -6 dB.

`B = fir1(N,Wn,'high')` designs an N'th order highpass filter.

You can also use `B = fir1(N,Wn,'low')` to design a lowpass filter.

If W_n is a two-element vector, $W_n = [W1 \ W2]$, `fir1` returns an order N bandpass filter with passband $W1 < W < W2$. You can also specify `B = fir1(N,Wn,'bandpass')`. If $W_n = [W1 \ W2]$, `B = fir1(N,Wn,'stop')` will design a bandstop filter.

►

Filtering

- ▶ The main thing about Matlab's filter design functions is that frequencies are specified relative to half the sampling frequency of the input data

```
>> help fir1
```

FIR filter design using the window method. `B = fir1(N,Wn)` designs an N'th order lowpass FIR digital filter and returns the filter coefficients in length N+1 vector B. The cut-off frequency Wn must be between $0 < Wn < 1.0$, with 1.0 corresponding to half the sample rate.

- ▶ `fir1` is the simplest FIR filter design method (`fir2` and `firls` are more sophisticated methods): Good for designing standard low pass, high pass and bandpass filters.
- ▶ For the exercise, you have to design all of the above three types of filters with the appropriate frequency limits
- ▶ Study the script **DSP/LowPassFilterDemo.m** to see how a low pass filter is designed and how it is applied (to the quadratic chirp signal)

Time frequency analysis

- ▶ Use Matlab's spectrogram function to make time-frequency plots of the signals that you have coded
 1. Generate signal time series with appropriate Nyquist sampling frequency
 2. Make spectrograms: When successful, [add spectrogram generation](#) to `test<functionName>.m` script
- ▶ See `DSP/SpecgrmQCDemo.m` for an example
- ▶ **Highly recommended:** See the documentation of `spectrogram` in Matlab (start with “help spectrogram” and follow up with the hyperlink at the end of the help)